

1. Administrative & Study Identification

Full Study Title: A prospective, multicenter, single arm study to evaluate the impact on the implementation of standardized hyperkalemia management in chronic kidney disease patients
Short Title/Acronym: HK-QIP Study (Hyperkalemia Quality Improvement Program)
Protocol Number: D9480L00029
NCT Number: NCT06884267
Sponsor Name: AstraZeneca
Investigational Product: Standard hyperkalemia management implementation (Guideline Determined Medical Therapy / GDMT)
Phase of Development: N/A for Quality Improvement
Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate whether quality improvement interventions and the implementation of standardized hyperkalemia management can improve the medical care process and clinical outcomes in chronic kidney disease (CKD) patients. * **Secondary Objectives:** To provide real-world evidence for standard hyperkalemia (HK) management with RAASi (Renin-Angiotensin-Aldosterone System inhibitor) optimization in Chinese non-dialysis chronic kidney disease (CKD-ND) patients.

2.2 Background & Rationale To address the gap in RAASi optimization in CKD-ND patients with $S_{K+} > 5.0$ mmol/L. HK is highly prevalent in patients with CKD, particularly those on RAASi, often leading to the detrimental discontinuation or down-titration of these essential therapies. This quality improvement study is being conducted to demonstrate that integrating standard clinical pathways and quality audits can drive GDMT adherence, improve S_{K+} monitoring, and enhance long-term patient outcomes.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional (Quality Improvement Study) * **Control Type:** Single-arm * **Blinding:** Open-label (None) * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target N :** 1,000 adult Chinese patients * **Number of Sites:** Around 50 sites in China * **Estimated Study Start:** 25-Jun-2025 * **Estimated Primary Completion:** 03-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years (Adult Chinese patients). 2. Confirmed diagnosis of non-dialysis chronic kidney disease (CKD-ND) with renal insufficiency. 3. Confirmed hyperkalemia defined as $\text{SK}^+ > 5.0 \text{ mmol/L}$.

4.2 Exclusion Criteria 1. Patients on dialysis. 2. Pseudohyperkalemia signs and symptoms (e.g., hemolyzed blood specimen due to excessive fist clenching, difficult/traumatic venipuncture, or history of severe leukocytosis or thrombocytosis). 3. Patients with acute kidney injury (AKI) or diabetic ketoacidosis (DKA).

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** RAASi targeted dose treatment and SK^+ testing at least once every 3 months (or once/month for high-risk HK patients, defined as CKD with DKD, HF, or RAASi initiation/up-titration). * **Route of Administration:** Oral (Standard disease management including RAASi use and prescription of potassium-lowering therapy). * **Duration of Treatment:** Up to 96 weeks of follow-up.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard concomitant medications for CKD, hypertension, heart failure (e.g., RAASi), and related co-morbidities as determined by the Health Care Professional (HCP). * **Prohibited:** Dialysis treatments (as they exclude the patient from the non-dialysis study population).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, SK^+ Labs
Baseline	Day 0	Patient Enrolment, HCP treatment choice, First Disease Education
Interim	Weeks 12, 24, 36, 48, 72	Onsite training, Efficacy Assessment, Data collection, Quality Audit tracking
End of Study	Week 96	Final Onsite Visit, Exit Interview, Final Data Collection

(Note: Remote visits take place every 4 weeks [Q4W] and patient daily checklists are reviewed every 12 weeks [Q12W])

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Proportion of patients achieving RAASi therapy optimization with normal SK^+ levels after 48 weeks of intervention. * **Measurement Method:** Clinical

and prescription records evaluation combined with central/local laboratory analysis for normal ≤ 5.0 mmol/L).

7.2 Secondary & Exploratory Endpoints * **Endpoint 1:** Achievement of ≤ 5.0 mmol/L, severe HK incidence, and time to first HK recurrence. * **Endpoint 2:** Cardiorenal benefits (including tracking of sustained eGFR decline, ESRD progression, HF worsening, and CV death).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection via remote Q4W contacts, onsite visits (Weeks 12, 24, 36, 48, 72, 96), and patient daily checklist reviews.
 - **Serious Adverse Events (SAEs):** Reported within typical 24 hours of site awareness.
 - **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by AstraZeneca.
-

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary effectiveness outcomes. Safety Set for all AE/SAE evaluations.
 - **Sample Size Justification:** Target enrollment of $N=1,000$ to detect meaningful shifts in GDMT adherence and RAASi optimization across ~50 sites.
 - **Handling of Missing Data:** Standard protocol handling (e.g., LOCF or Multiple Imputation) where applicable for continuous laboratory endpoints.
-

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Quality audits include tracking HCP's perception and action of standardized HK management (quarterly from the first patient enrolled at the site). Patient adherence to medication and ≤ 5.0 mmol/L monitoring evaluated via remote visit every 4 weeks and checklist reviews every 12 weeks.
 - **Audit Trail:** Quality audit results serve as feedback for additional medical education. Additional GDMT trainings are added at the individual site level (≤ 1 time/site quarterly, total ≤ 6 times/site) if the quality audit is off target during weeks 0-48.
-

11. Study Identifiers & Governance

- **Primary ID:** D9480L00029
 - **Registry Number:** NCT06884267
 - **Sponsor/Lead Organization:** AstraZeneca
 - **Study Title:** HK-QIP Study
 - **Official Regulatory Title:** A prospective, multicenter, single arm study to evaluate the impact on the implementation of standardized hyperkalemia management in chronic kidney disease patients
-

12. Clinical Framework (Hyperkalemia Specific)

- **Primary Condition:** Hyperkalemia in non-dialysis Chronic Kidney Disease (CKD-ND)
 - **Diagnosis Threshold:** $\text{K}^+ > 5.0 \text{ mmol/L}$
 - **Medical Methodology:** Guideline Determined Medical Therapy (GDMT)
 - **Optimization Target:** RAASi targeted dose treatment
-

13. Study Procedures & Methodology

- **Management Pathway:** Standardized clinical pathway based on guideline adoption for HK
 - **Education Protocol (HCP):** GDMT training 5 times at both study level and individual site level (with supplementary visits added if quality audits are off target)
 - **Education Protocol (Patient):** Onsite training 6 times during 0-96 weeks, combined with patient self-learning materials
 - **Quality Audit Mechanism:** Tracking HCP perception/action (quarterly) & tracking patient adherence to medication/monitoring
-

14. Observation & Follow-Up Schedule

- **Total Duration:** 96 Follow-up weeks
 - **Onsite Visit Cadence:** Weeks 12, 24, 36, 48, 72, 96
 - **Remote Monitoring Frequency:** Every 4 weeks (Q4W)
 - **Checklist Review Interval:** Every 12 weeks (Q12W)
-

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				